

# The Design of Safety Vectors on the Star Graph <sup>\*</sup>

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## Abstract

The concept of the safety vector can guide an efficient fault-tolerant routing on interconnection networks. The safety vector on the hypercube is based on the distance of a pair of nodes. However, the distance measure cannot be applied on the star graph directly since there are more than one routing path pattern on the star graph when the distances of two pairs of nodes are the same. In this paper, we first define the routing path pattern and path structure. And then, based on the definition, we propose the ways for constructing the safety vector of the star graph. The undirected safety vector of the star graph is a one-dimensional vector on each node. For increasing the accuracy of selecting routing paths, we further propose the directed safety vector, which is a two-dimensional vector on each node. The routing ability of our directed safety vector is superior to that of the safety level of the star graph. When we calculate the safety vector of one node, we have to access the safety vectors (storing the routing abilities of path patterns) of its neighboring nodes. Thus, we also propose a ranking method to rank the path patterns.

## 1 Introduction

In designing a multi-processor system, the topology of the communication structure is important. The topology can be represented as a graph, in which each node stands for a processor and each edge stands for a communication channel between a pair of processors. The Cayley graph [4] offers a group-theoretical model to study the symmetric interconnection networks such as the hypercube and the star graph. The *star graph* [1, 4] has good properties of the Cayley graph, and it has some better properties such as the degree and the diameter, compared with the hypercube.

When nodes or links fail in an interconnection network, some paths may not be available, and the routing may take more links or may not be routed at all. In such a situation, we need a fault-tolerant routing to maintain the availability of the system [2, 3, 5–7, 9, 13, 16–18, 20, 25]. Based on the information about faults, the fault-tolerant routing can be classified as follows [24]:

1. *Global information based method* [14]: Each node knows where all faults are, and the best possible path of any two nodes is known at first. One important issue in this method, however, is how to get the global information.
2. *Local information based method* [19]: Each node knows only the states of its neighboring nodes and links. The routing path between any two nodes is not known in

advance. In this method, we can apply the trial-and-error method like the depth-first search to find the routing paths. For keeping tracks, the message overhead is needed.

3. *Limited global information based method*: The information about faults around the neighborhood is gathered and processed so that a guide of the fault-distribution in the system is obtained. This guide is helpful to a routing, and the length of a path can be computed. The gathering work in this method is not as difficult as the global information based method. Moreover, there is no much message overhead. It is a compromised method of the above two methods.

There are two kinds of limited global information based methods. They are *safety level* [23] and *safety vector* (or *routing capability*) [24], both of which were studied on the hypercube. Generally, the safety vector can describe the fault-distribution more precisely.

Some fault-tolerant routing algorithms for the star graph have been presented [7, 9, 11, 15, 21]. The safety level applied on the star graph, however, does not perform well in fault tolerant routing [8]. Recently, Rezazad et al. proposed a fault-tolerant routing algorithm using the concept of unsafety vectors [22]. In this paper, we propose an efficient method for the safety vector on the star graph based on the limited global information based method, where the faulty nodes or links are given.

The rest of this paper is organized as follows. In Section 2, we review some properties and the safety level of the star graph. We also introduce the previous result of the safety vector on the hypercube. In Section 3, we propose the undirected safety vector of the star graph. For improving the routing ability of the undirected safety vector, we also propose the directed safety vector in Section 4. In Section 5, we present methods for computing the length of our safety vector and the rank of each routing path pattern. Finally, we conclude this paper in Section 6.

## 2 Preliminaries

In an  $n$ -dimensional star graph  $S_n$ , each node is identified by a distinct permutation of  $n$  symbols. For any two nodes  $u$  and  $v$  in  $S_n$ , if the identifier of  $u$  can be obtained from  $v$  by

swapping the first symbol with one another symbol of  $v$ , then there is a link connecting these two nodes, and  $u$  is a neighbor of  $v$  and vice versa. We use  $u^{(i)}$  to denote the neighbor of node  $u$  obtained by swapping the first symbol with the  $i$ th symbol of  $u$ .

*Routing* refers to sending a message from a *source* node to a *destination* node via some intermediate nodes in an interconnection network. The routing path should be as short as possible. According to the definition of the star graph, a routing path corresponds to a series of transpositions, and they can be represented by a permutation as a product of  $p$ -cycles, where  $p \geq 2$ . Then, the routing path can be found by these cycles from right to left. For example, from node 2143 to node 1234 in  $S_4$ , the permutation is  $\begin{pmatrix} 1234 \\ 2143 \end{pmatrix} = (12)(34) = (13)(14)(13)(12)$  and the routing path is  $2143 \rightarrow 1243 \rightarrow 4213 \rightarrow 3214 \rightarrow 1234$ .

The *distance* of two nodes  $u$  and  $v$  in a graph is the length of their shortest path. The *diameter* of a graph is the maximum of distances among all pairs of nodes in the graph. In  $S_n$ , the diameter is  $\left\lfloor \frac{3(n-1)}{2} \right\rfloor$  [1, 4].

There are some properties about the routing in the star graph. Let  $\pi$  be the permutation from the source to the destination, and  $c$  be the number of cycles of size not smaller than 2. Moreover, let ' $A$ ' denote the first symbol of the destination and  $m$  be the number of symbols of the source not at the correct positions of the destination. Then, the distance  $d(\pi)$  can be obtained by the following formula [1, 4].

$$d(\pi) = c + m - \begin{cases} 0, & \text{if } A \text{ is the same as the first symbol of the source,} \\ 2, & \text{otherwise.} \end{cases}$$

The neighbors on the shortest paths are called *preferred neighbors*, while the others are called *non-preferred neighbors*. In other words, for a source  $u$  and a destination  $v$ , if we move from  $u$  to one of its preferred neighbors, we are at a position closer to  $v$ . Let  $\Delta c$ ,  $\Delta m$ , and  $\Delta d$  denote the changes of  $c$ ,  $m$  and  $d(\pi)$ , respectively. Then, we can use these changes as fault-free routing rules to find the shortest paths. Table 1 shows the effect of moving a node to one of its neighbors, in which ' $X$ ' and ' $X$ ' are any two symbols of the destination different from ' $A$ '. In Table 1, a neighbor in case 1, 2, or 3 is a preferred neighbor.

The safety level represents the ability of performing a routing within some distance by the shortest path. In other words, if the safety level of node  $u$  is  $k$ , then  $u$  can route

Table 1: Fault-free routing rules.

| Case | First symbol of $u$ | Method                                                                         | New first symbol of $u^{(i)}$ | $\Delta c$ | $\Delta m$ | $\Delta d$ |
|------|---------------------|--------------------------------------------------------------------------------|-------------------------------|------------|------------|------------|
| 1    | $A$                 | Swap $A$ with any symbol $X$ not at the correct position                       | $X$                           | 0          | +1         | -1         |
| 2    | $X$                 | Swap $X$ to its correct position                                               | $A$<br>$X'$                   | -1<br>0    | -2<br>-1   | -1<br>-1   |
| 3    | $X$                 | Swap $X$ with any other symbol $X'$ whose position is in another cycle         | $X'$                          | -1         | 0          | -1         |
| 4    | $X$                 | Swap $X$ with any other symbol $X'$ or $A$ whose position is in the same cycle | $A$<br>$X'$                   | 0<br>+1    | -1<br>0    | +1<br>+1   |
| 5    | $A$<br>$X$          | Swap $X$ or $A$ with any other symbol $X'$ already in the correct position     | $X'$                          | +1<br>0    | +2<br>+1   | +1<br>+1   |

to any destination of distance  $k$  via the shortest path. In the star graph, the safety level of node  $u$  contains the safety level  $L_u[i]$  of each neighbor  $u^{(i)}$  of  $u$ ,  $2 \leq i \leq n$ . The safety levels of  $S_n$  can be determined as follows [8]:

**Step 1.** For each node  $u$ , let  $L_u[i] = 0$ ,  $2 \leq i \leq n$ .

**Step 2.** Do the following computation  $\left\lfloor \frac{3(n-1)}{2} \right\rfloor$  rounds:

For each node  $u$ , let

$$L_u[i] = \begin{cases} 0, & \text{if } u^{(i)} \text{ or } u \text{ is a faulty node, or} \\ & \text{if both } u \text{ and } u^{(i)} \text{ are two ends of a faulty link,} \\ \min\{L_{u^{(i)}}[j] | j \in \{2, 3, \dots, n\} - \{i\}\} + 1, & \text{otherwise,} \end{cases}$$

$2 \leq i \leq n$ .

Figure 1 shows an  $S_4$  with faulty nodes 1324 and 3412. In this example, after the first round,  $L_{3124} = (0, 1, 1)$ ; after the second round,  $L_{2134}[3] = \min\{L_{3124}[2], L_{3124}[4]\} + 1 = \min\{0, 1\} + 1 = 1$ . Table 2 illustrates the safety levels of the  $S_4$  in Figure 1. For a source  $u$  and a destination  $v$  with distance  $k$ , if there exists  $L_u[i] \geq k - 1$ ,  $2 \leq i \leq n$ , and  $u^{(i)}$  is a preferred neighbor, then there is a path of length  $k$  between  $u$  and  $v$ .

Although the safety level can keep the routing ability within some distance via the shortest path, it cannot be sure whether one shortest path exists beyond that distance. The safety vector, on the other hand, can keep the routing ability more precisely at some

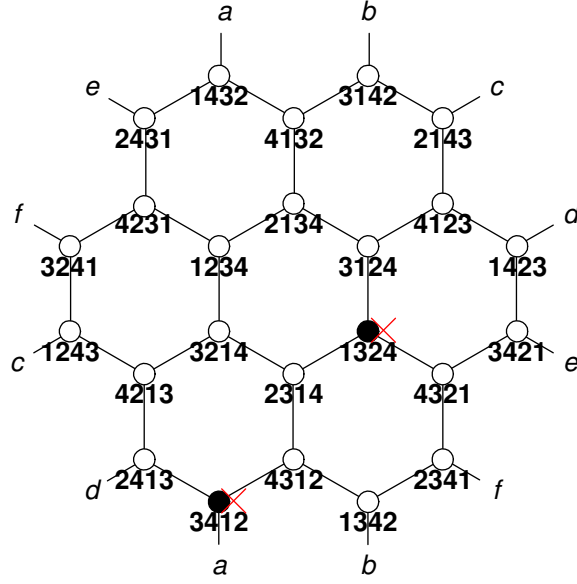


Figure 1:  $S_4$  with two faulty nodes 1324 and 3412.

Table 2: The safety levels of the  $S_4$  in Figure 1.

| node | safety level<br>$L[2, \dots, 4]$ | node | safety level<br>$L[2, \dots, 4]$ | node | safety level<br>$L[2, \dots, 4]$ | node | safety level<br>$L[2, \dots, 4]$ |
|------|----------------------------------|------|----------------------------------|------|----------------------------------|------|----------------------------------|
| 1234 | 223                              | 2134 | 312                              | 3124 | 033                              | 4123 | 241                              |
| 1243 | 323                              | 2143 | 323                              | 3142 | 223                              | 4132 | 132                              |
| 1324 | 000                              | 2314 | 301                              | 3214 | 132                              | 4213 | 142                              |
| 1342 | 312                              | 2341 | 412                              | 3241 | 233                              | 4231 | 233                              |
| 1423 | 212                              | 2413 | 330                              | 3412 | 000                              | 4312 | 031                              |
| 1432 | 303                              | 2431 | 421                              | 3421 | 122                              | 4321 | 330                              |

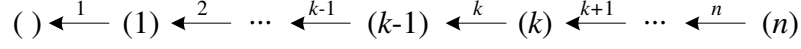


Figure 2: The path structure of  $Q_n$ .

exact distance. The safety vector of the hypercube is introduced below, which is helpful to understand our safety vector of the star graph.

In an  $n$ -dimensional hypercube  $Q_n$ , each node is identified by a distinct binary string of length  $n$ , and there is a link connecting two nodes if their binary strings differ exactly at one bit. Let  $u^{(i)}$  denote the neighbor of node  $u$  differing at the  $i$ th bit,  $1 \leq i \leq n$ . We say that node  $u$  is  $k$ -Hamming distance away from node  $v$  if there are  $k$  different bits between  $u$  and  $v$ .

In computing the safety vectors of the hypercube, if one node is an end of a faulty link, we consider the other end as a faulty node. The safety vector of node  $u$  in  $Q_n$  can be determined as follows [24]:

**Step 1.** If  $u$  is a faulty node, let  $u[k] = 0$ ,  $1 \leq k \leq n$ , and the safety vector of  $u$  is determined completely.

**Step 2.** If  $u$  is not a faulty node, let  $u[1] = \begin{cases} 0, & \text{if } u \text{ is an end of a faulty link,} \\ 1, & \text{otherwise.} \end{cases}$   
Then, for  $k = 2, 3, \dots, n$ , let  $u[k] = \begin{cases} 0, & \text{if } \sum_{1 \leq i \leq n} u^{(i)}[k-1] \leq n-k, \\ 1, & \text{otherwise.} \end{cases}$

For any one source node, there are  $n$  different classes of destinations. In each class, the distances between the source and the destinations are the same. The safety vector of node  $u$  in  $Q_n$  describes the routing abilities from  $u$  to the  $n$  classes of destinations [10,24]. We can find that a source node plays  $n$  kinds of *roles*, and each role corresponds to some destinations of a specific distance. For routing to one destination, the source has to route to one of its neighbors, and this neighbor becomes the new source. Different roles have different numbers of preferred neighbors. Therefore, the relations between roles and nodes can describe the routing abilities. We use a *path structure*, as shown in Figure 2, to represent the relations in the hypercube  $Q_n$ .

In the path structure,  $(k)$  represents the role of a source to the destinations of  $k$ -Hamming distance, and  $(k-1) \xleftarrow{k} (k)$  means that  $(k)$  has  $k$  preferred neighbors of role  $(k-1)$  that the source can route to the destination. The other  $n-k$  non-preferred neighbors are of role  $(k+1)$ . If one node has the  $(k)$  routing ability, it can surely route to any destination of  $k$ -Hamming distance via a shortest path. To compute the  $(k)$  routing ability of some node  $u$ , we have to check the  $(k-1)$  routing ability for all  $n$  neighbors of  $u$ . If there exists  $k$  nodes in the  $n$  neighbors of  $u$  such that all of these  $k$  nodes have the  $(k-1)$  routing ability, then  $u$  has the  $(k)$  routing ability, and we let  $u[k] = 1$ .

### 3 The undirected safety vector

As described in Section 2, a routing in the star graph can be represented by a product of  $p$ -cycles,  $p \geq 2$ . We call such a product structure a *pattern*. A pattern is represented by one 'A' and some 'X's, where 'A' and 'X' respectively denote the first symbol and any other symbol of the destination. Permutations in the same product structure belong to the identical pattern. For example, permutations  $\binom{1234}{2143} = (21)(34)$  and  $\binom{1234}{3412} = (31)(24)$  belong to pattern  $(XA)(XX)$ . A pattern plays a role of one source node. By the fault-free routing rules, permutations in the same pattern have similar shortest paths. However, the relations between roles and nodes cannot be described by the path structure of  $Q_n$ . For example, routing from nodes 2143, 3412, and 4321 to node 1234, the permutations are  $\binom{1234}{2143} = (21)(34)$ ,  $\binom{1234}{3412} = (31)(24)$ , and  $\binom{1234}{4321} = (41)(23)$ , respectively. They all belong to pattern  $(XA)(XX)$  and have distance 4. Their shortest paths are similar as shown in Figure 3. For another example, routing from nodes 1342 and 1423 to node 1234, the permutations are  $\binom{1234}{1342} = (234)$  and  $\binom{1234}{1423} = (243)$ , respectively. They both belong to pattern  $(XXX)$  and have the similar shortest paths of distance 4, as shown in Figure 4. We can observe that patterns on different shortest paths may be all different even when the distances are the same. Therefore, we need to find another path structure for the star graph  $S_n$ .

Using the fault-free routing rules, the relations between all patterns can be established. To obtain a pattern  $\alpha$ , we need distance  $d(\alpha)$  and the first symbol of the source. For



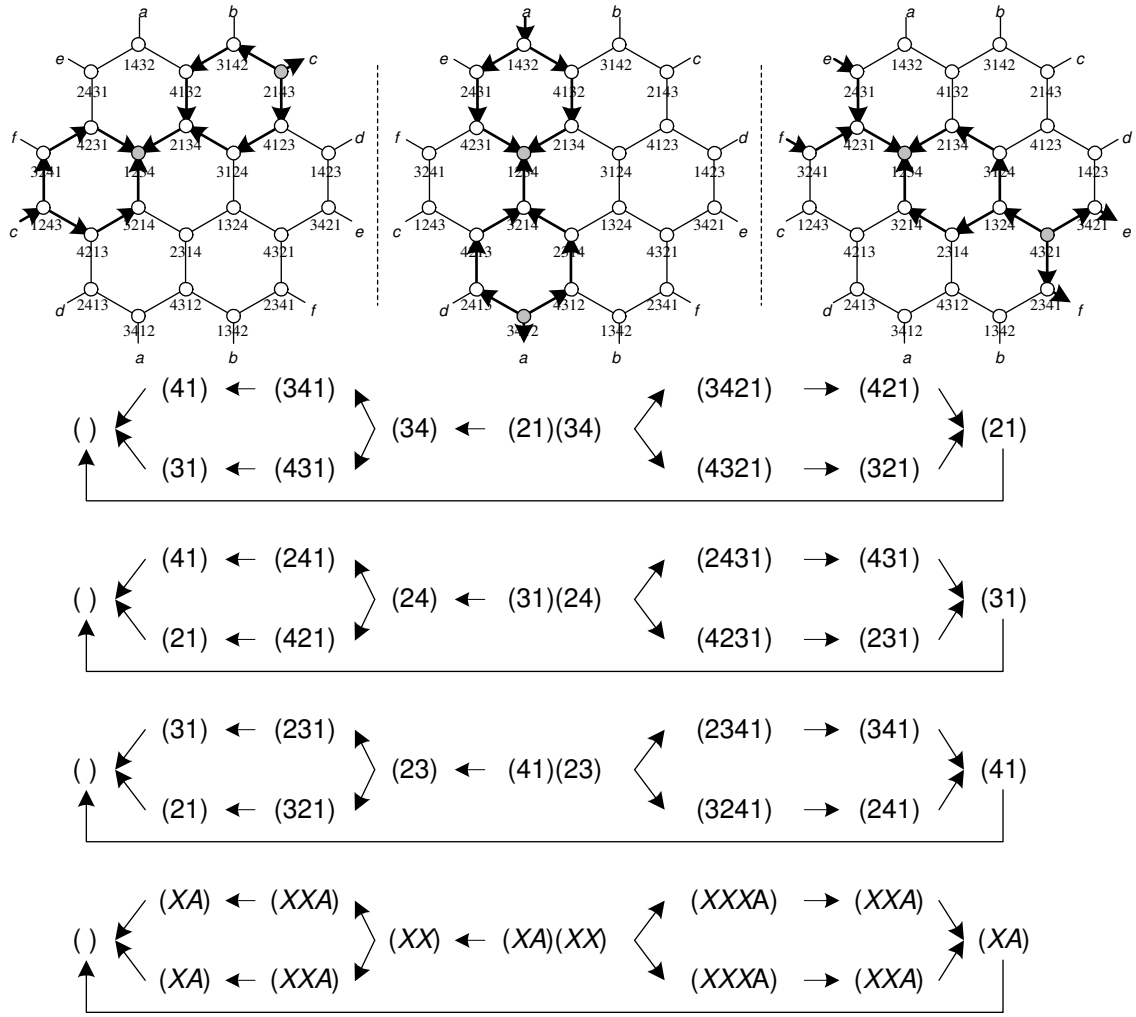


Figure 3: The shortest paths from source nodes in pattern  $(XA)(XX)$ .

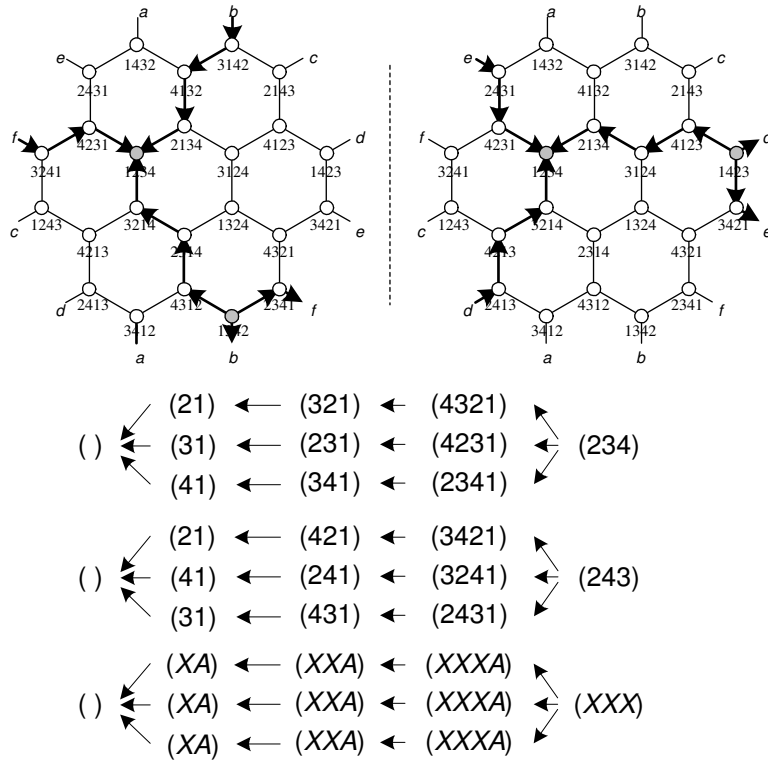


Figure 4: The shortest paths from source nodes in pattern  $(XXX)$ .

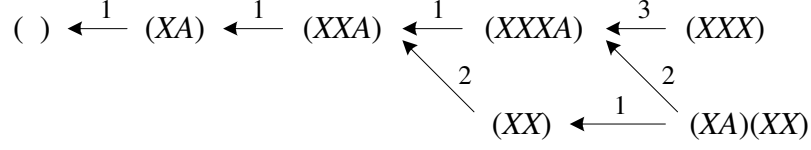


Figure 5: The path structure of  $S_4$ .

instance, in  $S_4$ , if  $d(\alpha) = 3$  and the first symbol of the source is 'A', then  $\alpha = (XX)$ . The reason is that  $m \geq 2c$  and  $c+m = 3$ , thus we obtain  $c = 1$ ,  $m = 2$ , and  $\alpha = (XX)$ . On the other hand, if  $d(\alpha) = 3$  and the first symbol of the source is not 'A', then  $\alpha = (XXXA)$ . This is because  $m \geq 2c$  and  $c + m - 2 = 3$ , then  $c = 1$ ,  $m = 4$ , and  $\alpha = (XXXA)$ . Similarly, if  $d(\alpha) = 4$ , then  $\alpha = (XXX)$  or  $\alpha = (XA)(XX)$ . With this argument, we can obtain all patterns of  $S_4$  as follows:  $( )$ ,  $(XA)$ ,  $(XXA)$ ,  $(XXXA)$ ,  $(XX)$ ,  $(XXX)$ , and  $(XA)(XX)$ , where  $( )$  means that the source is identical to the destination.

Continue discussing the patterns of  $S_4$ . For pattern  $(XXX)$ , if we swap the first symbol 'A' with any symbol not in the correct position, we have  $(XXXA) \xleftarrow{3} (XXX)$ ,  $\Delta c = 0$ ,  $\Delta m = 1$ , and the number 3 means that there are 3 preferred neighbors. For pattern  $(XA)(XX)$ , if we swap the first symbol 'X' with some symbol in another cycle, we have  $(XXXA) \xleftarrow{2} (XA)(XX)$ ,  $\Delta c = -1$ , and  $\Delta m = 0$ . In addition, if we swap the first symbol 'X' to its correct position, we obtain  $(XX) \xleftarrow{1} (XA)(XX)$ ,  $\Delta c = -1$ , and  $\Delta m = -2$ . By the above reasoning, the path structure of  $S_4$  can be established as shown in Figure 5.

With the path structure of Figure 5, we can find the shortest paths in  $S_4$ . For example, let  $(XA)(XX)$  be one role of a source  $u$  to a destination  $v$ . Since  $(XXXA) \xleftarrow{2} (XA)(XX)$  and  $(XX) \xleftarrow{1} (XA)(XX)$ ,  $u$  has three preferred neighbors to route to  $v$ : one is in pattern  $(XX)$  and the others are in pattern  $(XXXA)$ . Patterns  $(XX)$  and  $(XXXA)$  are called the *preferred patterns* of  $(XA)(XX)$ . In the star graph, for one source  $u$  in pattern  $\alpha$ , the preferred patterns of  $\alpha$  are the patterns that the preferred neighbors of  $u$  belong to. If  $u$  has the routing ability of pattern  $\alpha$ , there is at least one shortest path from  $u$  to the destinations via some preferred patterns of  $\alpha$ .

Let  $p(\alpha)$  be the set of  $\alpha$ 's preferred patterns; i.e. if  $\alpha' \in p(\alpha)$ , then  $d(\alpha') = d(\alpha) - 1$ . Moreover, let  $n(\alpha, \alpha')$  denote the number of preferred neighbors (in pattern  $\alpha'$ ) of a node (in pattern  $\alpha$ ). For example, if  $\alpha = (XA)(XX)$  in  $S_4$ , then  $d(\alpha) = 4$ ,  $p(\alpha) = \{(XXXXA), (XX)\}$ ,  $n(\alpha, (XXXXA)) = 2$ , and  $n(\alpha, (XX)) = 1$ . Let  $|\Omega|$  denote the number of elements in a set  $\Omega$ . If one node is an end of a faulty link, we consider the other end as a faulty node. Let  $\alpha$  be a pattern and  $\Omega_{u,\alpha} = \{i | 2 \leq i \leq n, u^{(i)}[\alpha] = 0\}$ . The undirected safety vector of one node  $u$  in  $S_n$  can be defined and determined as follows.

**Step 1.** If  $u$  is a faulty node, let  $u[\alpha] = 0$  for each  $\alpha$ , and the undirected safety vector of  $u$  is determined completely.

**Step 2.** If  $u$  is not a faulty node, let

$$u[(XA)] = \begin{cases} 0, & \text{if } u \text{ is an end of a faulty link,} \\ 1, & \text{otherwise.} \end{cases}$$

Then, for each  $\alpha$ ,  $\alpha \neq (XA)$ , let

$$u[\alpha] = \begin{cases} 1, & \text{if } |\cup_{\alpha' \in p(\alpha)} \Omega_{u,\alpha'}| < \sum_{\alpha' \in p(\alpha)} n(\alpha, \alpha'), \\ 0, & \text{otherwise.} \end{cases}$$

Note that the notation  $\cup$  means the set union operation. To obtain  $u[\alpha]$ , we have to compute  $u^{(i)}[\alpha']$  and let  $d(\alpha) = d(\alpha') + 1$  from all neighbors of  $u$ , where  $2 \leq i \leq n$  and  $\alpha' \in p(\alpha)$ . At first, we compute the routing ability of distance 1, then compute the routing abilities of distances  $2, 3, \dots, \lfloor \frac{3(n-1)}{2} \rfloor$ . Therefore, the safety vectors are obtained iteratively. Table 3 lists the undirected safety vectors of the star graph in Figure 1.

To set  $u[\alpha] = 1$ ,  $\alpha \neq (XA)$ , we have the following two cases, which are illustrated.

**Case I.**  $\exists \alpha' \in p(\alpha)$ ,  $|\Omega_{u,\alpha'}| < n(\alpha, \alpha')$ .

Consider the routing ability of  $\alpha = (XA)(XX)$  for a source  $u = 2134$ . The preferred patterns of  $\alpha$  are  $(XXXXA)$  and  $(XX)$ . For preferred pattern  $(XXXXA)$ , the preferred neighbors of  $u$  are  $2134^{(4)} = 4132$  and  $2134^{(3)} = 3124$ . From Table 3, we can see  $|\Omega_{u,(XXXXA)}| = |\{3, 4\}| = 2 = n(\alpha, (XXXXA))$ , since the routing abilities of  $(XXXXA)$  for both nodes 4132 and 3124 are 0. This means that the preferred neighbors 4132 and 3124 do not have the corresponding routing ability (the corresponding shortest paths) for  $\alpha$ . For the other preferred pattern  $(XX)$ , the preferred neighbor of  $u$  is  $2134^{(2)} = 1234$ . From Table 3,  $|\Omega_{u,(XX)}| = |\{\}| = 0 < n(\alpha, (XX)) = 1$ . The

Table 3: The undirected safety vectors of the  $S_4$  in Figure 1.

| node | $(XA)$ | $(XXA)$ | $(XXXA)$ | $(XX)$ | $(XXX)$ | $(XA)(XX)$ |
|------|--------|---------|----------|--------|---------|------------|
| 1234 | 1      | 1       | 1        | 1      | 1       | 1          |
| 1243 | 1      | 1       | 1        | 1      | 1       | 1          |
| 1324 | 0      | 0       | 0        | 0      | 0       | 0          |
| 1342 | 1      | 1       | 0        | 1      | 1       | 1          |
| 1423 | 1      | 1       | 0        | 1      | 0       | 1          |
| 1432 | 1      | 0       | 0        | 1      | 0       | 0          |
| 2134 | 1      | 1       | 0        | 1      | 1       | 1          |
| 2143 | 1      | 1       | 1        | 1      | 1       | 1          |
| 2314 | 1      | 0       | 0        | 0      | 0       | 0          |
| 2341 | 1      | 1       | 0        | 1      | 1       | 1          |
| 2413 | 1      | 0       | 0        | 1      | 0       | 0          |
| 2431 | 1      | 1       | 0        | 1      | 1       | 1          |
| 3124 | 1      | 0       | 0        | 1      | 0       | 0          |
| 3142 | 1      | 1       | 1        | 1      | 1       | 1          |
| 3214 | 1      | 1       | 0        | 1      | 1       | 1          |
| 3241 | 1      | 1       | 1        | 1      | 1       | 1          |
| 3412 | 0      | 0       | 0        | 0      | 0       | 0          |
| 3421 | 1      | 1       | 0        | 1      | 1       | 1          |
| 4123 | 1      | 1       | 0        | 1      | 1       | 1          |
| 4132 | 1      | 1       | 0        | 1      | 1       | 1          |
| 4213 | 1      | 1       | 0        | 1      | 1       | 1          |
| 4231 | 1      | 1       | 1        | 1      | 1       | 1          |
| 4312 | 1      | 0       | 0        | 0      | 0       | 0          |
| 4321 | 1      | 0       | 0        | 1      | 1       | 0          |

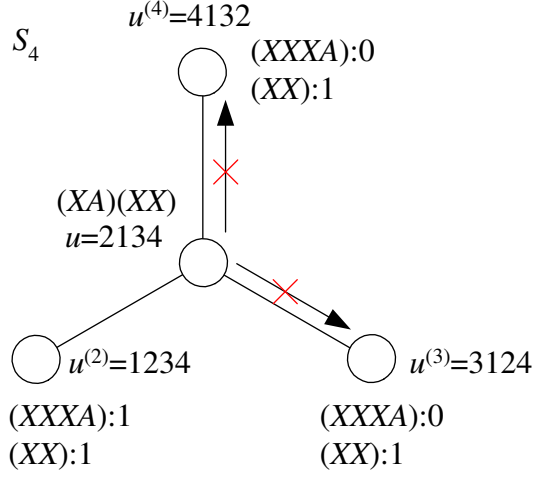


Figure 6: Computing the safety vector of node 2134 with  $\alpha = (XA)(XX)$ .

routing ability of  $(XX)$  for node 1234 is 1, then the preferred neighbor 1234 has the corresponding routing ability for  $\alpha$  and we can set  $2134[(XA)(XX)] = 1$ . Destinations such as nodes 1243, 4312, and 3421 can be routed via the shortest paths from node 2134 because node 2134 plays the role of  $(XA)(XX)$  to these destinations and  $2134[(XA)(XX)] = 1$ . Figure 6 illustrates this example.

**Case II.**  $(\forall \alpha' \in p(\alpha), |\Omega_{u,\alpha'}| = n(\alpha, \alpha'))$  and  $|\cup_{\alpha' \in p(\alpha)} \Omega_{u,\alpha'}| < \sum_{\alpha' \in p(\alpha)} n(\alpha, \alpha')$ .

Consider the routing ability of  $\alpha = (XA)(XX)$  for a source  $u = 3214$ . For preferred pattern  $(XXXXA)$ , the preferred neighbors of  $u$  are  $3214^{(4)} = 4213$  and  $3214^{(2)} = 2314$ . From Table 3,  $|\Omega_{u,(XXXXA)}| = |\{2, 4\}| = 2 = n(\alpha, (XXXXA))$ . The routing abilities of  $(XXXXA)$  for both nodes 4213 and 2314 are 0. For the other preferred pattern  $(XX)$ , the preferred neighbor of  $u$  is  $3214^{(3)} = 1234$ . From Table 3,  $|\Omega_{u,(XX)}| = |\{2\}| = 1 = n(\alpha, (XX))$ . However,  $|\Omega_{u,(XXXXA)} \cup \Omega_{u,(XX)}| = |\{2, 4\}| = 2 < n(\alpha, (XXXXA)) + n(\alpha, (XX)) = 2 + 1 = 3$ . The routing ability of  $(XX)$  for node 1234 is 1. Thus, the preferred neighbor 1234 has the corresponding routing ability for  $\alpha$ , and we can set  $3214[(XA)(XX)] = 1$ . A destination such as node 1432 can be routed via the shortest path from node 3214 because node 3214 plays the role of  $(XA)(XX)$  to destination 1432 and  $3214[(XA)(XX)] = 1$ . Figure 7 illustrates this

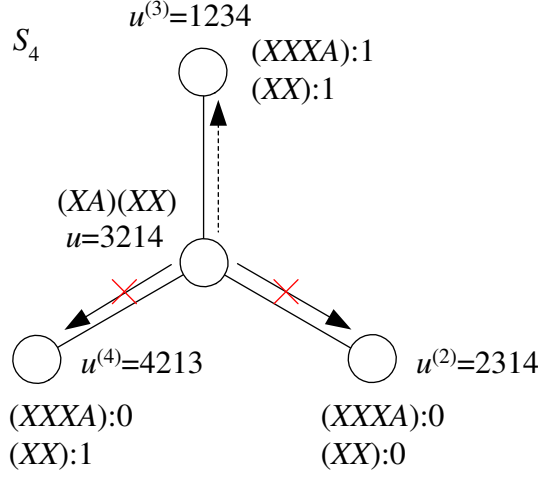


Figure 7: Computing the safety vector of node 3214 with  $\alpha = (XA)(XX)$ .

example.

We have some properties of the undirected safety vector of  $S_n$  as follows.

**Property 1** For one node  $u$ , if the nearest faulty node is  $k$ -distance away from  $u$ , then  $u[\alpha] = 1$  for each pattern  $\alpha$ , where  $d(\alpha) < k$ .

**Theorem 1** Let  $u$  be a source node in pattern  $\alpha$  with  $d(\alpha) = k$  to a destination  $v$ . If  $u^{(i)}$  is a preferred neighbor having  $u^{(i)}[\alpha'] = 1$  and  $\alpha' \in p(\alpha)$ , then there is a shortest path of length  $k$  from  $u$  to  $v$  via  $u^{(i)}$ .

**Proof:** Initially, if  $u^{(i)}$  is an end of a faulty link or  $u^{(i)}$  is a faulty node, then  $u^{(i)}[(XA)] = 0$ ; otherwise,  $u^{(i)}[(XA)] = 1$ . It is trivial that this theorem holds if  $\alpha = (XXXA)$ . We shall complete the proof by induction on  $d(\alpha)$ . Suppose this theorem also holds for each pattern  $\alpha$  with  $d(\alpha) \leq k - 1$ . For each pattern  $\alpha$  with  $d(\alpha) = k$ , if  $u^{(i)}$  is a preferred neighbor in pattern  $\alpha'$  having  $u^{(i)}[\alpha'] = 1$  and  $d(\alpha') = k - 1$ , there exists some  $j$  such that  $(u^{(i)})^{(j)}[\alpha''] = 1$ ,  $\alpha'' \in p(\alpha')$ , and node  $(u^{(i)})^{(j)}$  is a preferred neighbor of  $u^{(i)}$  to route to  $v$ . Therefore, there is a shortest path from  $u$  to  $v$  via  $u^{(i)}$  and  $(u^{(i)})^{(j)}$ . This completes the proof.  $\square$

**Corollary 1** *Let  $u$  be a source node in pattern  $\alpha'$  with  $d(\alpha') = k$  to a destination  $v$ . If  $u^{(i)}$  is a non-preferred neighbor having  $u^{(i)}[\alpha] = 1$  and  $\alpha' \in p(\alpha)$ , then there is a non-shortest path of length  $k + 2$  from  $u$  to  $v$  via  $u^{(i)}$ .*

## 4 The directed safety vector

In the undirected safety vector, some situations may affect the routing abilities. For example, in Figure 8, the faulty link  $(2314, 4312)$  yields  $2314[(XA)] = 0$ . Suppose the source node  $u$  is 2314 and the destination node  $v$  is 2134. For node 1324 with  $\alpha = (XXA)$ ,  $(XA)$  is the only preferred pattern. Since  $1324^{(3)}[(XA)] = 2314[(XA)] = 0$ , we have  $|\Omega_{(1324),(XA)}| = |\{3\}| = 1$ . Similarly,  $|\Omega_{(3214),(XA)}| = |\{2\}| = 1$ . Because  $n((XXA), (XA)) = 1$ , we obtain that the routing abilities of  $(XXA)$  for nodes 1324 and 3214 are 0, as shown in Figure 9. Since the pattern from nodes 1324 and 3214 to node 2134 is  $(XXA)$ , the routing abilities of  $(XXA)$  for nodes  $u^{(3)} = 1324$  and  $u^{(2)} = 3214$  conclude that the shortest path between  $u$  and  $v$  may not exist. But actually, there are two shortest paths from  $u$  to  $v$ . This is because setting the routing abilities of  $(XXA)$  for  $u^{(2)}$  and  $u^{(3)}$  has included the routing ability for  $u$ . A situation affecting any routing abilities is called an *illusion*. Because illusions in routing do not follow the shortest paths, we should not involve these illusions in any shortest path routing.

For excluding illusions, we propose the *directed safety vector*, in which each node  $u$  stores the safety vectors of all its neighbors. In other words,  $u[i]$  stores the directed safety vector of  $u^{(i)}$ ,  $2 \leq i \leq n$ . When we compute the directed safety vector  $u[i]$ , we do not involve any illusion from  $u^{(i)}$  to  $u$ . Therefore, the directed safety vectors describes the fault-distribution more precisely. Let  $\alpha$  be a pattern and  $\Omega_{u^{(i)}, \alpha} = \{j | 2 \leq j \leq n, u^{(i)}[j][\alpha] = 0\}$ . The directed safety vector of one node  $u$  in  $S_n$  can be determined as follows.

**Step 1.** Initially, if  $\alpha = ()$ , then  $u[i][()] = 1$ ,  $2 \leq i \leq n$ .

**Step 2.** If either  $u$  is a faulty node or  $u$  is an end of a faulty link, let  $u[i][\alpha] = 0, \forall \alpha$ .

**Step 3.** If neither  $u$  is not a faulty node nor  $u$  is not an end of a faulty link,  $\forall \alpha \neq ()$  let



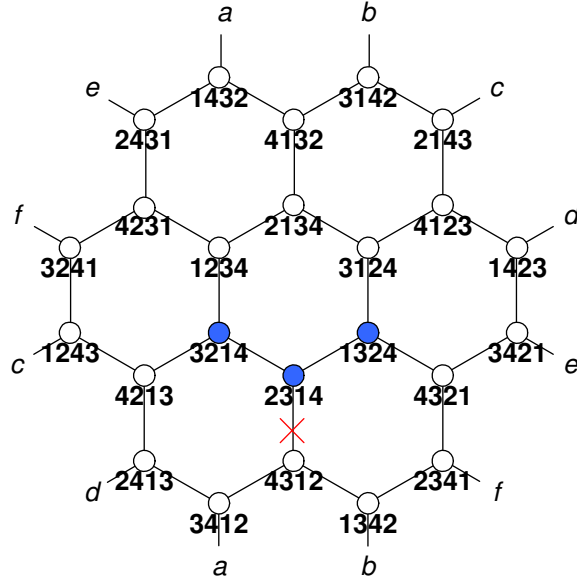


Figure 8:  $S_4$  with the faulty link (2314,4312).

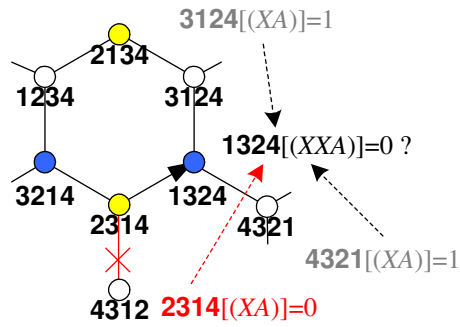


Figure 9: The influence of the illusion in Figure 8.

$$u[i][\alpha] = \begin{cases} 1, & \text{if } |\cup_{\alpha' \in p(\alpha)} (\Omega_{u(i), \alpha'} - \{i\})| < (\sum_{\alpha' \in p(\alpha)} n(\alpha, \alpha')) - 1, \\ 0, & \text{otherwise.} \end{cases}$$

To obtain  $u[i][\alpha]$ , the routing ability of distance 1 is computed at first, then the routing abilities of distances 2, 3,  $\dots$ ,  $\left\lfloor \frac{3(n-1)}{2} \right\rfloor$  are computed, iteratively. Table 4 lists the directed safety vectors of the  $S_4$  in Figure 1.

Table 4: The directed safety vectors of the  $S_4$  in Figure 1.

| node | neighbor | () | (XA) | (XXA) | (XXXA) | (XX) | (XXX) | (XA)(XX) |
|------|----------|----|------|-------|--------|------|-------|----------|
| 1234 | 2134     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 3214     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 4231     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
| 1243 | 2143     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 3241     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 4213     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
| 1324 | 2314     |    |      |       |        |      |       |          |
|      | 3124     |    |      |       |        |      |       |          |
|      | 4231     |    |      |       |        |      |       |          |
| 1342 | 2341     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
|      | 3142     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 4312     | 1  | 1    | 0     | 0      | 0    | 0     | 0        |
| 1423 | 2413     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
|      | 3421     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 4123     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
| 1432 | 2431     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 3412     | 1  | 0    | 0     | 0      | 0    | 0     | 0        |
|      | 4132     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
| 2134 | 1234     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 3124     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
|      | 4132     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
| 2143 | 1243     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 3142     | 1  | 1    | 1     | 1      | 1    | 0     | 0        |
|      | 4123     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
| 2314 | 1324     | 1  | 0    | 0     | 0      | 0    | 0     | 0        |
|      | 3214     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 4312     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
| 2341 | 1342     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
|      | 3241     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 4321     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
| 2413 | 1423     | 1  | 1    | 1     | 1      | 1    | 0     | 0        |
|      | 3412     | 1  | 0    | 0     | 0      | 0    | 0     | 0        |
|      | 4213     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |

to be continued

continued

| node | neighbor | () | (XA) | (XXA) | (XXXA) | (XX) | (XXX) | (XA)(XX) |
|------|----------|----|------|-------|--------|------|-------|----------|
| 2431 | 1432     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
|      | 3421     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 4231     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
| 3124 | 1324     | 1  | 0    | 0     | 0      | 0    | 0     | 0        |
|      | 2134     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 4123     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
| 3142 | 1342     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 2143     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 4132     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
| 3214 | 1234     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 2314     | 1  | 1    | 0     | 0      | 0    | 0     | 0        |
|      | 4213     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
| 3241 | 1243     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 2341     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 4231     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
| 3412 | 1432     |    |      |       |        |      |       |          |
|      | 2413     |    |      |       |        |      |       |          |
|      | 4312     |    |      |       |        |      |       |          |
| 3421 | 1423     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 2431     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
|      | 4321     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
| 4123 | 1423     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 2143     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 3124     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
| 4132 | 1432     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
|      | 2134     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
|      | 3142     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
| 4213 | 1243     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 2413     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
|      | 3214     | 1  | 1    | 1     | 0      | 0    | 1     | 1        |
| 4231 | 1234     | 1  | 1    | 1     | 1      | 1    | 0     | 0        |
|      | 2431     | 1  | 1    | 1     | 0      | 0    | 0     | 0        |
|      | 3241     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
| 4312 | 1342     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 2314     | 1  | 1    | 0     | 0      | 0    | 1     | 1        |
|      | 3412     | 1  | 0    | 0     | 0      | 0    | 0     | 0        |
| 4321 | 1324     | 1  | 0    | 0     | 0      | 0    | 0     | 0        |
|      | 2341     | 1  | 1    | 1     | 1      | 1    | 1     | 1        |
|      | 3421     | 1  | 1    | 1     | 1      | 1    | 0     | 0        |

The directed safety vector has the following results.

**Property 2** For one node  $u$ , if the nearest faulty node is  $k$ -distance away from  $u^{(i)}$ , then  $u[i][\alpha] = 1$  for each pattern  $\alpha$ , where  $d(\alpha) < k$ .

**Theorem 2** Let  $u$  be a source node in pattern  $\alpha$  with  $d(\alpha) = k$  to a destination  $v$ . If  $u^{(i)}$  is a preferred neighbor having  $u[i][\alpha'] = 1$  and  $\alpha' \in p(\alpha)$ , then there is a shortest path of length  $k$  from  $u$  to  $v$  via  $u^{(i)}$ .

**Proof:** The proof of this theorem is similar to that of Theorem 1 and it is omitted here.  $\square$

**Corollary 2** Let  $u$  be a source node in pattern  $\alpha'$  with  $d(\alpha') = k$  to a destination  $v$ . If  $u^{(i)}$  is a non-preferred neighbor having  $u[i][\alpha] = 1$  and  $\alpha' \in p(\alpha)$ , then there is a non-shortest path of length  $k + 2$  from  $u$  to  $v$  via  $u^{(i)}$ .

Since our directed safety vector divides each node into more than one status according to the path patterns and the safety level [8] only concentrates each node's neighbors into one value, our directed safety vector is more powerful than the safety level. In other words, if  $u[i][\alpha] = 1$  for all patterns  $\alpha$  with  $d(\alpha) \leq k$ , then  $L_u[i] \leq k$ , where  $k$  is a constant and  $L_u[i]$  is the safety level of  $u$  for  $u^{(i)}$ .

## 5 The length and the rank of a safety vector

To calculate the safety vector of one node, we have to access the values of safety vectors of its neighbors. We need an indexing scheme to pick up the value in the vector (array) we desire. Thus, we develop a ranking method to rank the path patterns.

We first calculate the length of a safety vector, which is the number of all considered patterns. It can be obtained from the number of partitions of non-negative integer  $k$ , denoted by  $\phi_k$  [12]. Let  $P_n$  be the number of all possible patterns in  $S_n$ . Then,  $P_n = \sum_{k=1}^n \phi_{n-k}$ , where  $\phi_{n-k}$  corresponds to all possible patterns beginning with  $\underbrace{(X \cdots X A)}_k$ . For example,  $\phi_4 = 5$  since 4 can be partitioned into (4), (3+1), (2+2), (2+1+1), and (1+1+1+1). Thus, in  $S_7$ ,  $\phi_4 = \phi_{7-3}$  corresponds to patterns  $\underbrace{(X X A)}_3 \underbrace{(X X X X)}_4, \underbrace{(X X A)}_3 \underbrace{(X X X)}_3 \underbrace{(X)}_1$ ,

Table 5: Ranking all patterns of  $S_7$ .

| Rank | Pattern         | Rank | Pattern         | Rank | Pattern           |
|------|-----------------|------|-----------------|------|-------------------|
| 1    | $(XXXXXXA)$     | 11   | $(XXA)(XX)$     | 21   | $(A)(XXXXX)$      |
| 2    | $(XXXXXA)$      | 12   | $(XXA)$         | 22   | $(A)(XXXX)(XX)$   |
| 3    | $(XXXXA)(XX)$   | 13   | $(XA)(XXXXX)$   | 23   | $(A)(XXX)$        |
| 4    | $(XXXXA)$       | 14   | $(XA)(XXXX)$    | 24   | $(A)(XXX)(XXX)$   |
| 5    | $(XXXA)(XXX)$   | 15   | $(XA)(XXX)(XX)$ | 25   | $(A)(XXX)(XX)$    |
| 6    | $(XXXA)(XX)$    | 16   | $(XA)(XXX)$     | 26   | $(A)(XXX)$        |
| 7    | $(XXXA)$        | 17   | $(XA)(XX)(XX)$  | 27   | $(A)(XX)(XX)(XX)$ |
| 8    | $(XXA)(XXXX)$   | 18   | $(XA)(XX)$      | 28   | $(A)(XX)(XX)$     |
| 9    | $(XXA)(XXX)$    | 19   | $(XA)$          | 29   | $(A)(XX)$         |
| 10   | $(XXA)(XX)(XX)$ | 20   | $(A)(XXXXXX)$   | 30   | $(A)$             |

$(\underbrace{XXX}_3)(\underbrace{XX}_2)(\underbrace{XX}_2)$ ,  $(\underbrace{XXX}_3)(\underbrace{XX}_2)(\underbrace{X}_1)(\underbrace{X}_1)$ , and  $(\underbrace{XXA}_3)(\underbrace{X}_1)(\underbrace{X}_1)(\underbrace{X}_1)(\underbrace{X}_1)$ . Note that symbol  $(X)$  can be omitted in the patterns; therefore,  $(\underbrace{XXX}_3)(\underbrace{XX}_2)(\underbrace{X}_1)(\underbrace{X}_1) = (XXXA)(XX)$ .

When the length becomes larger, it is getting difficult to look up some pattern in a safety vector. Thus, we need a ranking method for all patterns so that each pattern is associated with a distinct position number. Let  $\phi_{k,q}$  be the number of partitions of a non-negative integer  $k$  such that any element in each partition is less than  $q$ . For example,  $\phi_{4,3} = 3$  because it can be partitioned into  $(2+2)$ ,  $(2+1+1)$  and  $(1+1+1+1)$ . We rank patterns of  $S_n$  by the following rules:

**Rule 1.** Each pattern  $\alpha$  is in the format of  $\alpha = \underbrace{(X \cdots XA)}_{k_1} \underbrace{(X \cdots X)}_{k_2} \cdots \underbrace{(X \cdots X)}_{k_{C_\alpha}}$ ,  
 $k_2 \geq k_3 \geq \cdots \geq k_{C_\alpha}$ .

**Rule 2.** For two patterns  $\alpha$  and  $\beta$ , where  $\alpha = \underbrace{(X \cdots XA)}_{k_1} \underbrace{(X \cdots X)}_{k_2} \cdots \underbrace{(X \cdots X)}_{k_{C_\alpha}}$  and  $\beta = \underbrace{(X \cdots XA)}_{h_1} \underbrace{(X \cdots X)}_{h_2} \cdots \underbrace{(X \cdots X)}_{h_{C_\beta}}$ , the rank of  $\alpha$  is larger than the rank of  $\beta$  if  $k_1 k_2 \cdots k_{C_\alpha} < h_1 h_2 \cdots h_{C_\beta}$  in lexicographic order of strings.

For example, the rank of  $(XA)(XX)$  is larger than the rank of  $(XA)(XXX)$ . Table 5 shows the ranks of all patterns in  $S_7$ .

The rank of one pattern can be computed by the following theorem.

**Theorem 3** Let  $\alpha = \underbrace{(X \cdots XA)}_{k_1} \underbrace{(X \cdots X)}_{k_2} \cdots \underbrace{(X \cdots X)}_{k_C}$  be a pattern, where  $k_2 \geq k_3 \geq \cdots \geq k_C$ . Then, the rank of  $\alpha$  is  $\sum_{k=k_1+1}^n \phi_{n-k} + \sum_{i=2}^C \sum_{k=k_i+1}^{n-k_1-\cdots-k_{i-1}} \phi_{n-k_1-\cdots-k_{i-1}-k,k} + 1$ .

**Proof:** To complete this proof, we first compute the number of all possible patterns beginning with  $\underbrace{(X \cdots XA)}_k$ ,  $k \geq k_1 + 1$ , which can be obviously calculated as follows.

$$\phi_{n-(k_1+1)} + \phi_{n-(k_1+2)} + \cdots + \phi_{n-n} = \sum_{k=k_1+1}^n \phi_{n-k}.$$

Then, we compute the number of all possible patterns beginning with  $\underbrace{(X \cdots XA)}_{k_1} \underbrace{(X \cdots X)}_k$ ,  $k \geq k_2 + 1$ , which

$$\begin{aligned} & \phi_{(n-k_1)-(k_2+1),(k_2+1)} + \phi_{(n-k_1)-(k_2+2),(k_2+2)} + \cdots + \phi_{(n-k_1)-(n-k_1),(n-k_1)} \\ &= \sum_{k=k_2+1}^{n-k_1} \phi_{n-k_1-k,k}. \end{aligned}$$

Similarly, we compute the number of all possible patterns beginning with  $\underbrace{(X \cdots XA)}_{k_1} \underbrace{(X \cdots X)}_{k_2} \underbrace{(X \cdots X)}_k$ ,  $k \geq k_3 + 1$ , and we get

$$\begin{aligned} & \phi_{(n-k_1-k_2)-(k_3+1),(k_3+1)} + \phi_{(n-k_1-k_2)-(k_3+2),(k_3+2)} + \cdots + \phi_{(n-k_1-k_2)-(n-k_1-k_2),(n-k_1-k_2)} \\ &= \sum_{k=k_3+1}^{n-k_1-k_2} \phi_{n-k_1-k_2-k,k}. \end{aligned}$$

With the above argument, we can conclude that the theorem holds.  $\square$

In  $S_7$ , for example, the rank of  $(XA)(XXX)(XX)$  is

$$\begin{aligned} & \sum_{k=2+1}^7 \phi_{7-k} + \sum_{k=3+1}^{7-2} \phi_{7-2-k,k} + \sum_{k=2+1}^{7-2-3} \phi_{7-2-3-k,k} + 1 \\ &= (\phi_4 + \phi_3 + \phi_2 + \phi_1 + \phi_0) + (\phi_{1,4} + \phi_{0,5}) + 1 \\ &= (5 + 3 + 2 + 1 + 1) + (1 + 1) + 1 \\ &= 15. \end{aligned}$$

## 6 Conclusion

In this paper, we define the safety vector of the star graph based on the routing path patterns. We first present the *undirected safety vector*, which is a one-dimensional vector on each node. The basic property of the undirected safety vector is that if a routing ability is 1, there surely exists one shortest path from the source node to all corresponding destinations. But if the routing ability is 0, it is not sure whether there exists such a shortest path.

A good design for the safety vector is to increase the number of 1's in the vector so that it is close to the optimal number. However, to get the optimal safety vector is very difficult. For improving the undirected safety vector, we then propose the *directed safety vector*, which is a two-dimensional vector on each node that stores the routing capability of all its neighboring nodes. Our directed safety vector outperforms the safety level in judging the routing paths. But the safety vector requires more space and more time-consuming process than the safety level. It is a trade-off.

As mentioned above, if the routing ability is 0, it is not sure for the existence of the shortest path. We may apply the fuzzy concept for showing the routing abilities. The *statistical safety vector* may be appropriate to increase the accuracy of the routing abilities. In this method, the value of each element in the safety vector of node  $u$  is set between 0 and 1, not only 0 or 1, to represent the probability of a successful routing from  $u$  to all corresponding destinations. The statistical method should be worthy of further study.

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